

Course Name:	Digital Signal and Image Processing	Semester:	VI
Date of Performance:	18 / 02 / 2026	DIV/ Batch No:	A1
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Title: Implement the following point processing techniques in spatial domain:

- Image Negative.
- Thresholding.
- Gray level slicing with and without background
- Bit plane slicing
- Contrast stretching

Objective: To learn & understand point processing techniques.

Expected Outcome of Experiment:

CO	Outcome
CO4	Apply various spatial and frequency domain enhancement techniques for 1-D and 2-D signals.

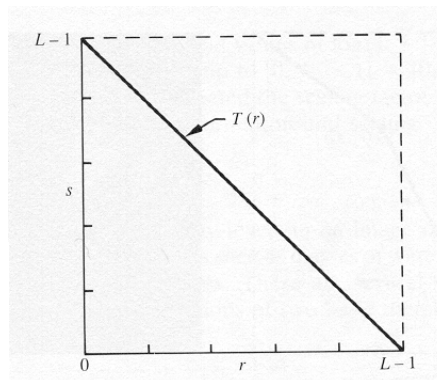
Books/ Journals/ Websites referred:

1. <http://www.mathworks.com/support/>
2. www.math.mtu.edu/~msgocken/intro/intro.html.
3. R. C.Gonsales R.E.Woods, "Digital Image Processing", Second edition, Pearson Education
4. S.Jayaraman, S Esakkirajan, T Veerakumar "Digital Image Processing "Mc Graw Hill.
5. S.Sridhar,"Digital Image processing", oxford university press, 1st edition."

Pre Lab/ Prior Concepts:

Image Negative:

Negative images are useful for enhancing white or grey detail embedded in dark regions of an image. Image negatives are obtained by using the transformation function $s=T(r)$.

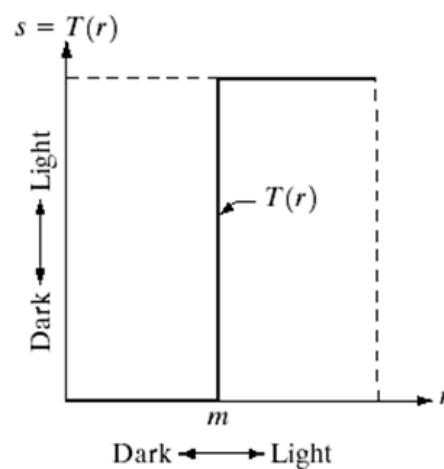


$[0, L-1]$ is the range of gray levels

$$S = L-1-r$$

Thresholding

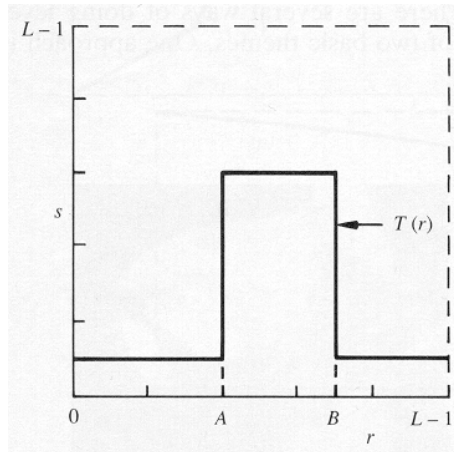
From a grayscale image, thresholding can be used to create binary images. The simplest thresholding methods replace each pixel in an image with a black pixel if the image intensity is less than some fixed constant T or a white pixel if the image intensity is



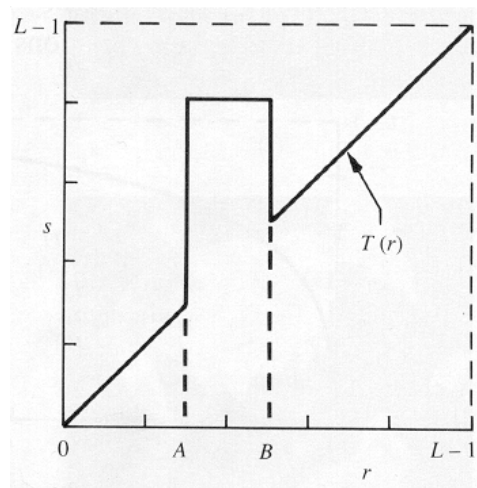
greater than that constant.

Gray Level Slicing

To highlight a specific range of gray levels in an image (e.g. to enhance certain features). One way is to display a high value for all gray levels in the range of interest and a low value for all other gray levels (binary image).

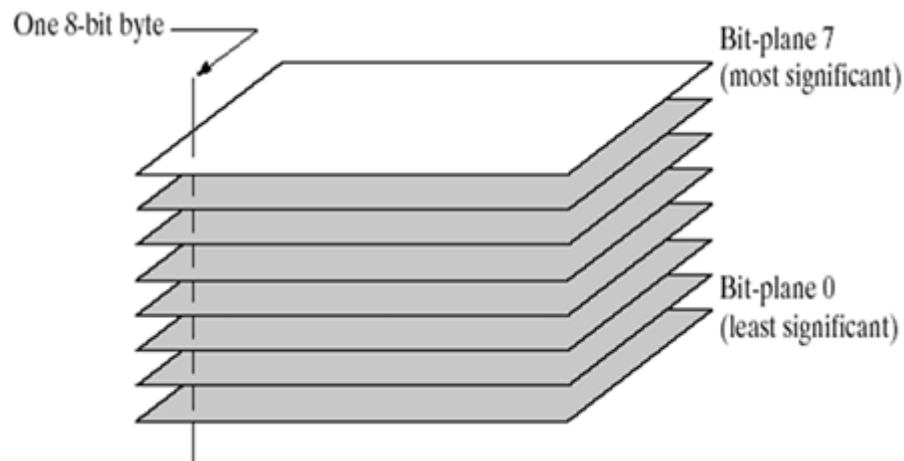


The second approach is to brighten the desired range of gray levels but preserve the background and gray-level tonalities in the image:



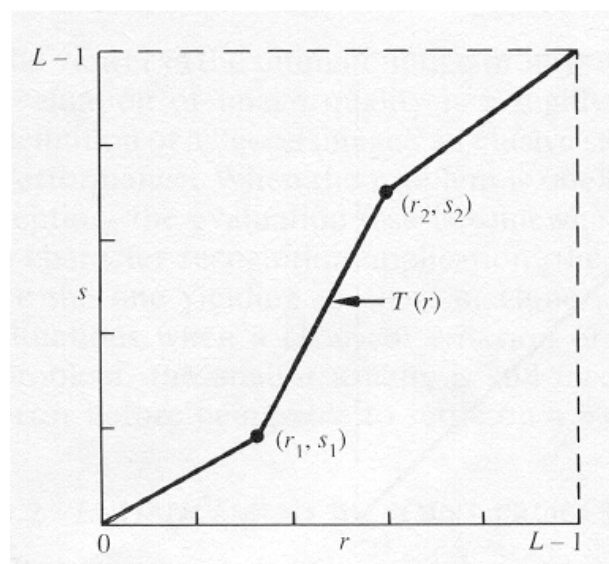
Bit plane slicing

Bit plane slicing is used to highlight the contribution made to the total image appearance by specific bits. Assuming that each pixel is represented by 8 bits, the image is composed of 8 1-bit planes. Plane 0 contains the least significant bit and plane 7 contains the most significant bit. Only the higher order bits (top four) contain visually significant data. The other bit planes contribute the more subtle details. Plane 7 corresponds exactly with an image thresholded at gray level 128.



Contrast stretching

Contrast stretching (often called normalization) is a simple image enhancement technique that attempts to improve the contrast in an image by 'stretching' the range of intensity values it contains to span a desired range of values, *e.g.* the the full range of pixel values that the image type concerned allows. It differs from the more sophisticated histogram equalization in that it can only apply a *linear* scaling function to the image pixel values. As a result the 'enhancement' is less harsh.



The locations of (r_1, s_1) and (r_2, s_2) control the shape of the transformation function.

- If $r_1 = s_1$ and $r_2 = s_2$ the transformation is a linear function and produces no changes.

- If $r_1=r_2$, $s_1=0$ and $s_2=L-1$, the transformation becomes a thresholding function that creates a binary image.
- Intermediate values of (r_1, s_1) and (r_2, s_2) produce various degrees of spread in the gray levels of the output image, thus affecting its contrast.

Generally, $r_1 \leq r_2$ and $s_1 \leq s_2$ is assumed.

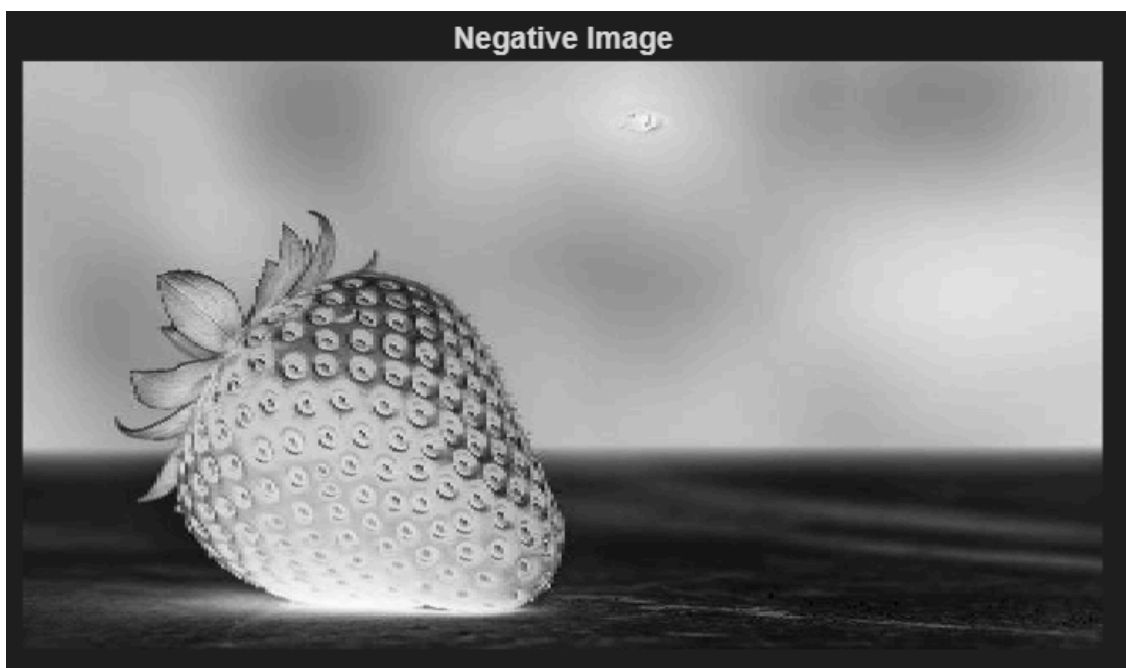
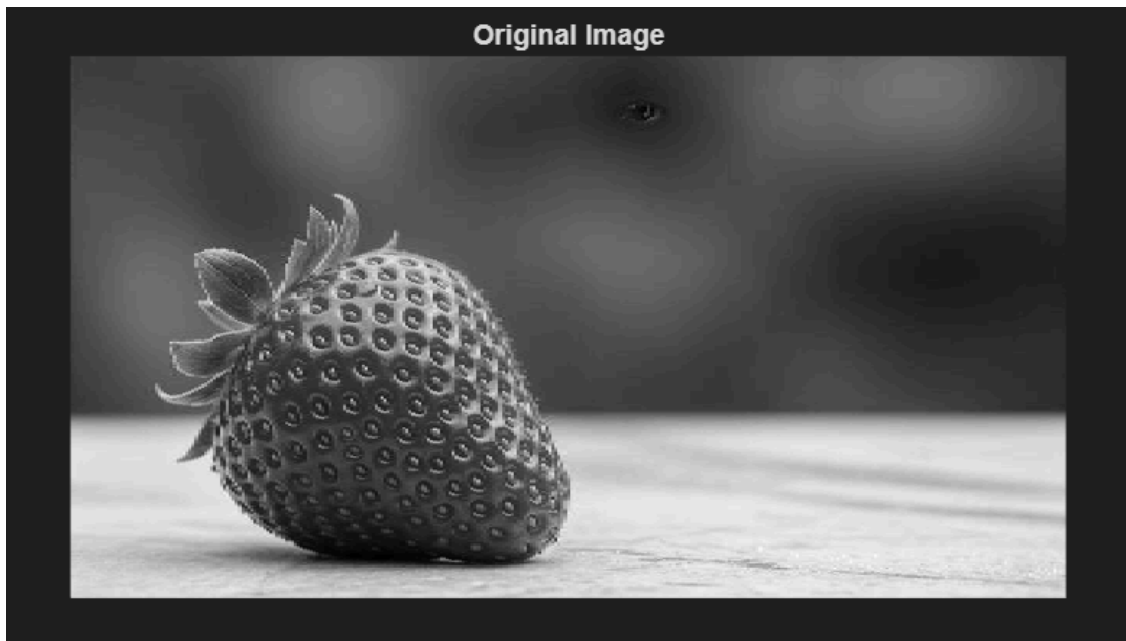
Implementation steps with screenshots:



Image Negative:

```
uiimport("/MATLAB Drive/image.jpg");  
  
image = imread("/MATLAB Drive/image.jpg");  
  
img = imread('image.jpg');  
  
if size(img,3) == 3  
  
    img = rgb2gray(img);  
  
end  
  
figure;  
  
imshow(img);  
  
title('Original Image');  
  
img_double = double(img);
```

```
L = 256;  
negative_img = uint8(L - 1 - img_double);  
figure;  
imshow(negative_img);  
title('Negative Image');
```



Thresholding:

```
clc;

clear;

close all;

img = imread('image.jpg');

if size(img,3) == 3

    img = rgb2gray(img);

end

figure;

imshow(img);

title('Original Grayscale Image');


T_low = 50;  % Very low threshold

T_high = 205;  % Very high threshold


% Apply Thresholding

binary_low = uint8(255 * (img >= T_low));

binary_high = uint8(255 * (img >= T_high));

figure;

imshow(binary_low);

title('Extreme Thresholding - Low T = 50');

figure;

imshow(binary_high);

title('Extreme Thresholding - High T = 205');
```



Gray Level Slicing:

```
clc;
```

```
clear;
```

```
close all;
```



```
img = imread('image.jpg');

if size(img,3) == 3
    img = rgb2gray(img);
end

figure;
imshow(img);
title('Original Grayscale Image');

A = 100;
B = 180;

binary_slice = uint8(255 * (img >= A & img <= B));

figure;
imshow(binary_slice);
title('Gray Level Slicing - Binary');

preserve_slice = img; % Start with original
preserve_slice(img >= A & img <= B) = 255;

figure;
imshow(preserve_slice);
title('Gray Level Slicing - Background Preserved');
```



Bit plane slicing:

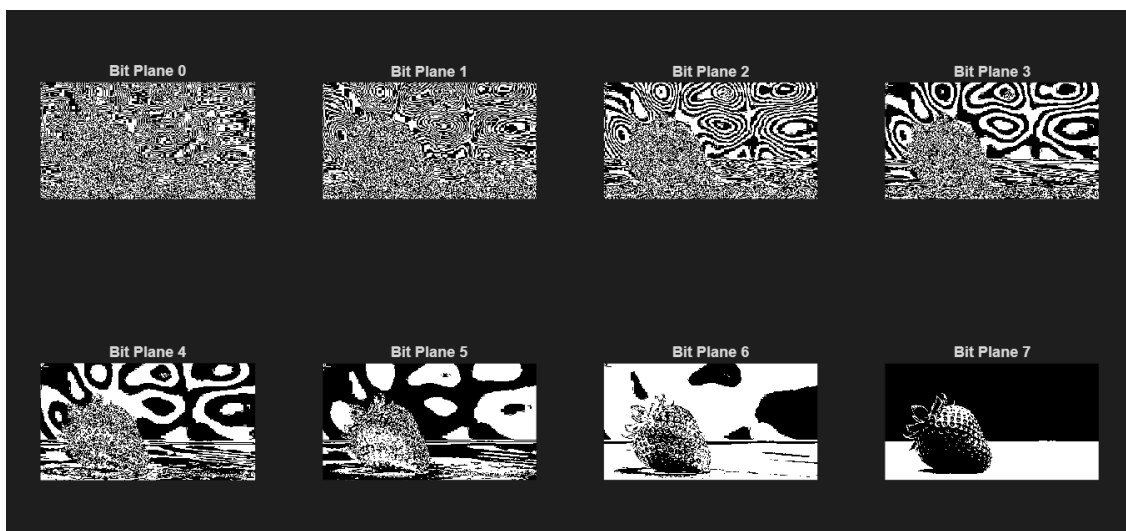
```
clc;
```

```
clear;
```

```
close all;
```

```
img = imread('image.jpg');
```

```
if size(img,3) == 3  
  
    img = rgb2gray(img);  
  
end  
  
figure;  
  
imshow(img);  
  
title('Original Grayscale Image');  
  
figure;  
  
for k = 0:7  
  
    bit_plane = bitget(img, k+1);  
  
  
    subplot(2,4,k+1);  
  
    imshow(bit_plane);  
  
    title(['Bit Plane ', num2str(k)]);  
  
end
```



Contrast stretching:

```
clc;
```

```
clear;

close all;

img = imread('image.jpg');

if size(img,3) == 3

    img = rgb2gray(img);

end

img = double(img);

L = 256;

figure;

imshow(uint8(img));

title('Original Image');


r1 = 70;

s1 = 20;

r2 = 180;

s2 = 240;


stretched = zeros(size(img));

for i = 1:size(img,1)

    for j = 1:size(img,2)

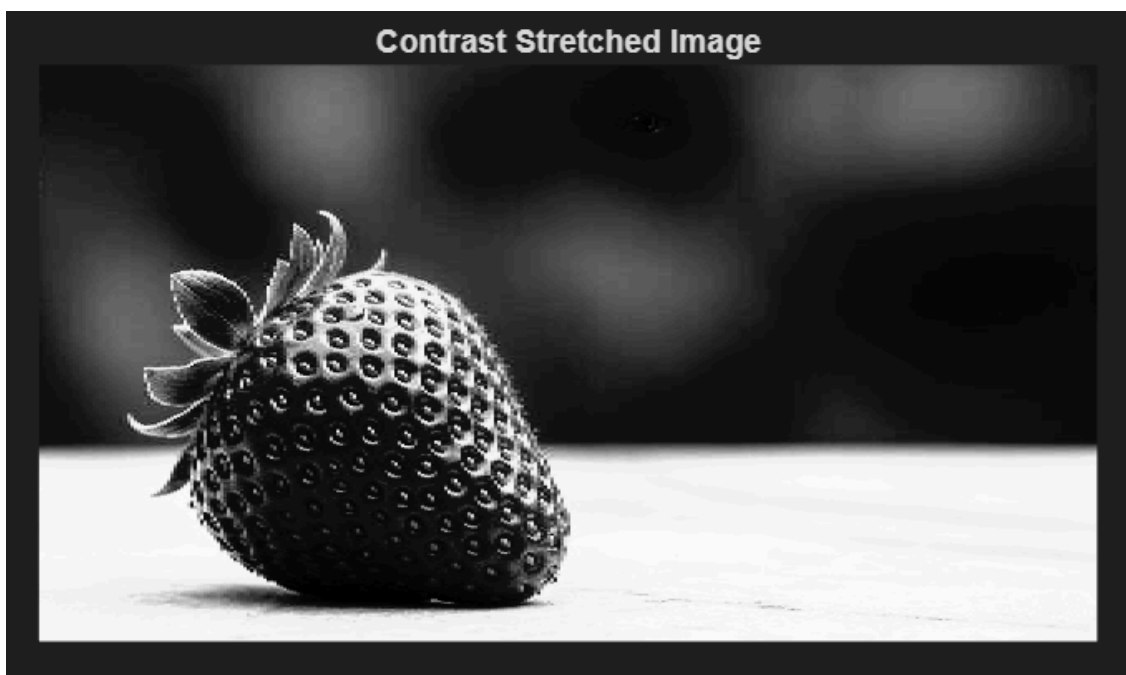
        r = img(i,j);

        if r < r1

            stretched(i,j) = (s1/r1)*r;

        elseif r <= r2
```

```
stretched(i,j) = ((s2-s1)/(r2-r1))*(r-r1) + s1;  
  
else  
  
stretched(i,j) = ((L-1-s2)/(L-1-r2))*(r-r2) + s2;  
  
end  
  
end  
  
stretched = uint8(stretched);  
  
figure;  
  
imshow(stretched);  
  
title('Contrast Stretched Image');
```



Conclusion:

I successfully implemented various point processing techniques in the spatial domain, including image negative, thresholding, gray level slicing, bit plane slicing, and contrast stretching using MATLAB. I observed how pixel-wise intensity transformations enhance image features, highlight specific details, and improve overall visual quality.

Date: 18/02/2026

Signature of faculty in-charge

Post Lab Descriptive Questions

- 1. Explain the role of bit plane slicing in achieving Steganography concept.**
Bit plane slicing helps separate an image into different binary layers. In steganography, secret data can be embedded in the lower-order bit planes (LSBs) because changes there are visually insignificant. This allows hidden information to be stored without noticeably altering the image appearance.
- 2. Explain the use of gray level slicing**
Gray level slicing is used to highlight specific intensity ranges in an image. It enhances features of interest while either suppressing or preserving the background. This technique is useful in medical imaging, satellite imagery, and pattern detection where particular structures must be emphasized.

Post Lab Objective Questions

- 1. Thresholding function in contrast stretching creates**
 - a) binary image
 - b) high quality image
 - c) enhanced image
 - d) low quality image
- 2. When is the contrast stretching transformation a linear function, for r and s as gray-value of image before and after processing respectively?**
 - a) $r_1 = s_1$ and $r_2 = s_2$
 - b) $r_1 = r_2, s_1 = 0$ and $s_2 = L - 1$, L is the max gray value allowed
 - c) $r_1 = 1$ and $r_2 = 0$
 - d) None of the mentioned
- 3. Which gray-level transformation increase the dynamic range of gray-level in the image?**
 - a) Power-law transformations
 - b) Negative transformations
 - c) Contrast stretching
 - d) None of the mentioned
- 4. When is the contrast stretching transformation a thresholding function, for r and s as gray-value of image before and after processing respectively?**

- a) $r1 = s1$ and $r2 = s2$
- b) $r1 = r2, s1 = 0$ and $s2 = L - 1$, L is the max gray value allowed
- c) $r1 = 1$ and $r2 = 0$
- d) None of the mentioned

5. What condition prevents the intensity artifacts to be created while processing with contrast stretching, if r and s are gray-values of image before and after processing respectively?

- a) $r1 = s1$ and $r2 = s2$
- b) $r1 = r2, s1 = 0$ and $s2 = L - 1$, L is the max gray value allowed
- c) $r1 = 1$ and $r2 = 0$
- d) $r1 \leq r2$ and $s1 \leq s2$